

An optimization problem is a problem of form

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, p\end{array}$$

where  $x \in \mathbb{R}^n$  is called the *optimization variable* and the function  $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$  the *objective function*. The inequalities  $f_i(x) \leq 0$  and equalities  $h_j(x) = 0$  are called inequality and equality constraints respectively.

The set  $\mathcal{D}$ , defined by

$$\mathcal{D} = \bigcap_{i=1}^m \text{dom } f_i \cap \bigcap_{j=1}^p \text{dom } h_j,$$

is called the *domain* of the optimization problem.

The *optimal value*  $p^*$  of the problem is defined as

$$p^* = \inf\{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m; h_j(x) = 0, j = 1, \dots, p\}.$$

To note,  $p^*$  is allowed to take on the extended values  $\pm\infty$ .

**0.0.1 See also**

**0.0.2 References**